Design and Simulation of a 1×2 Silicon Photonic Power Splitter

SiEPIC EBeam PDK Layout, Verification, and Circuit-Level Simulation

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Abstract

This report presents the layout implementation and circuit-level simulation of a 1×2 silicon photonic power splitter using KLayout, SiEPIC-Tools, and the EBeam process design kit. The circuit integrates one input grating coupler, a Y-branch splitter, two output grating couplers, and routed silicon waveguides. The completed layout was checked using design-rule and SiEPIC functional verification, including connectivity, component overlap, and device-recognition checks. The simulated transmission spectra of the two output channels closely overlap from 1500 to 1600 nm and reach a maximum near 1550 nm, demonstrating balanced two-channel splitting and correct circuit connectivity.

***Index Terms*—** silicon photonics, Y-branch, optical power splitter, grating coupler, SiEPIC, EBeam PDK, KLayout, INTERCONNECT.

I. INTRODUCTION

Silicon photonic integrated circuits combine optical routing and functional devices on a compact chip platform. A 1×2 splitter is a fundamental building block used to distribute an input optical signal into two branches for routing, monitoring, interferometers, and larger photonic networks. The objective of this work was to complete an end-to-end PIC workflow: place library components, route waveguides, verify that the layout is physically and functionally valid, and simulate the circuit response.

The design was implemented with the SiEPIC EBeam component library. The central functional element is a Y-branch that divides the input signal into two nominally equal paths. Grating couplers provide the external optical interfaces, while waveguide PCells connect the components and retain the DevRec and PinRec information required by the SiEPIC verification flow.

II. CIRCUIT COMPONENTS AND FUNCTION

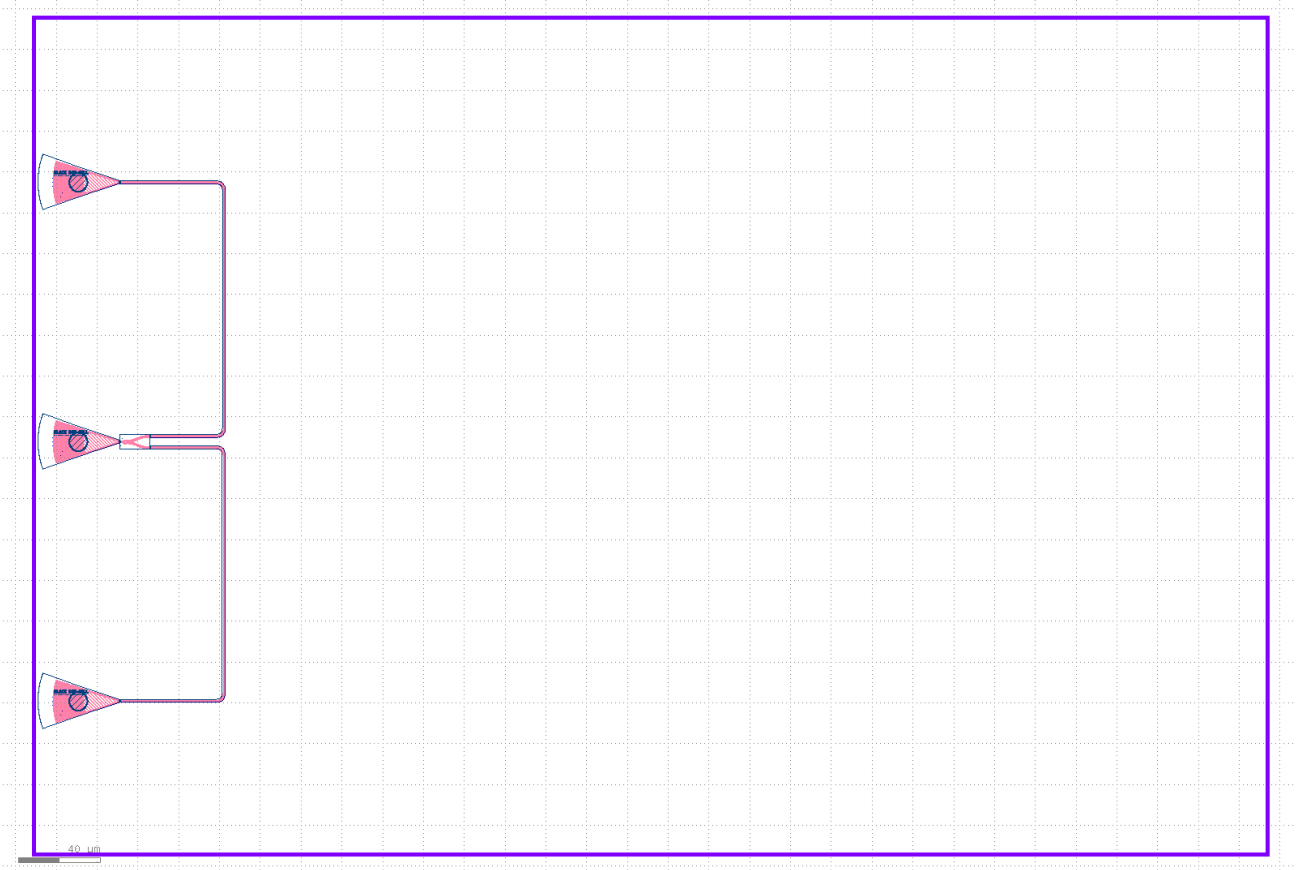
Element	Purpose in the Test Circuit
Input grating coupler	Couples light between an external optical fiber and the on-chip silicon waveguide. In the simulated test configuration, this port excites the splitter input.
1×2 Y-branch splitter	Divides the guided input field into two output branches. For an ideal symmetric splitter, each branch receives one-half of the input power, corresponding to a 3 dB splitting term before other losses.
Output grating couplers (2)	Provide two separate output ports so that the transmission spectrum of each branch can be measured and compared.
Silicon waveguides	Route light between the grating couplers and Y-branch. Smooth bends are used to connect the vertically arranged couplers while maintaining optical continuity.
DevRec and PinRec metadata	Define component boundaries and optical pins for automated verification. These records enable checks for disconnected pins, overlapping components, invalid pins, and shapes outside recognized components.

III. LAYOUT IMPLEMENTATION AND VERIFICATION

The layout uses three grating couplers arranged along the left side of the die area. The middle grating coupler acts as the input and is connected directly to the Y-branch. The upper and lower Y-branch outputs are routed to the

corresponding output grating couplers using waveguides with rounded 90-degree routing sections. The purple rectangular boundary indicates the overall design area.

Fig. 1. Final KLayout implementation of the 1×2 splitter test circuit.



The completed design was evaluated with the EBeam design-rule check and SiEPIC automated verification. During layout debugging, duplicate waveguide components and an isolated polygon outside any DevRec region were removed. The corrected design passed the local verification flow, confirming that the final hierarchy contains recognized components without the reported overlap or out-of-component shape errors.

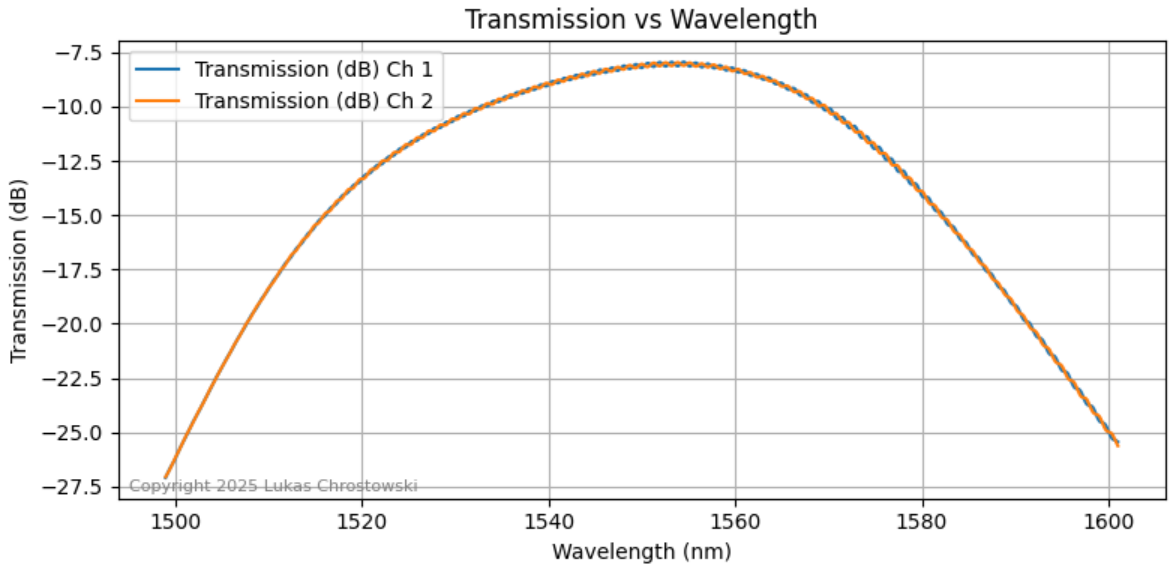
IV. TEST METHOD AND EXPECTED FUNCTION

A wavelength sweep from 1500 to 1600 nm was used to evaluate the optical transmission from the input port to the two output channels. The test has three primary functions:

1. **Connectivity check:** A nonzero spectral response at both outputs confirms that the input grating coupler, Y-branch, routed waveguides, and output ports are connected in the circuit model.
2. **Splitting-balance check:** The difference between Channel 1 and Channel 2 indicates the balance of the 1×2 splitter and the symmetry of the two routed branches.
3. **Spectral-response check:** The wavelength-dependent curve identifies the useful operating region and the wavelength of maximum end-to-end transmission.

V. SIMULATION RESULTS AND DISCUSSION

Fig. 2. Simulated transmission spectra of Channel 1 and Channel 2 from 1500 to 1600 nm.



The two simulated traces are nearly coincident across the full wavelength range, indicating that the two output paths have closely matched transmission. The response rises from approximately −26 dB near 1500 nm to a maximum of about −8 dB around 1555 nm, then decreases toward approximately −25 dB at 1600 nm. The close overlap of the blue and orange traces demonstrates balanced operation of the splitter within the resolution visible in the plotted result.

The plotted transmission represents the complete modeled path rather than only the intrinsic Y-branch division. Therefore, the approximately −8 dB peak should not be interpreted as the splitter excess loss alone. The result may include the nominal 3 dB power division together with grating-coupler response, waveguide propagation and bend losses, and any model-dependent insertion losses. A dedicated splitter-loss extraction would require normalization to a reference structure and comparison of the summed output powers with the input power.

TABLE I. SUMMARY OF OBSERVED PERFORMANCE

Metric	Observed Result
Sweep range	1500–1600 nm
Approximate peak wavelength	Approximately 1555 nm
Approximate peak transmission	Approximately −8 dB for both channels
Channel matching	The two traces are visually almost identical
Functional conclusion	Balanced 1×2 splitting and valid end-to-end connectivity

VI. CONCLUSION

A complete 1×2 silicon photonic power-splitter circuit was successfully implemented and simulated using the SiEPIC EBeam workflow. The design integrates one input grating coupler, a Y-branch, two output grating couplers, and routed silicon waveguides. Layout verification was completed after resolving component-overlap and stray-shape issues. The simulated spectra show two closely matched output channels with maximum transmission near 1555 nm, confirming the intended balanced splitting function and correct circuit connectivity.

REFERENCES

[1] SiEPIC-Tools, silicon photonics design, verification, and simulation workflow.
[2] SiEPIC EBeam Process Design Kit, component library and design rules.
[3] KLayout, layout editing and design-rule-check environment.
[4] Ansys Lumerical INTERCONNECT, photonic integrated circuit simulation environment.